



Week 03 Task-07

Footprinting & OSINT (Passive Reconnaissance)

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1) Executive Summary

Footprinting and reconnaissance is the foundational step in real-world ethical hacking. Footprinting through OSINT (**Open Source Intelligence**) is the process of collecting information about the target without directly interacting with the target systems. Passive reconnaissance allows security professionals to gather as much detail about the target as possible without being detected. Examples include domain information, DNS records, subdomains, and emails of the company's employees. This information serves as the building block for threat actors to plan the attack according to the online presence of an individual or presence of organizations.

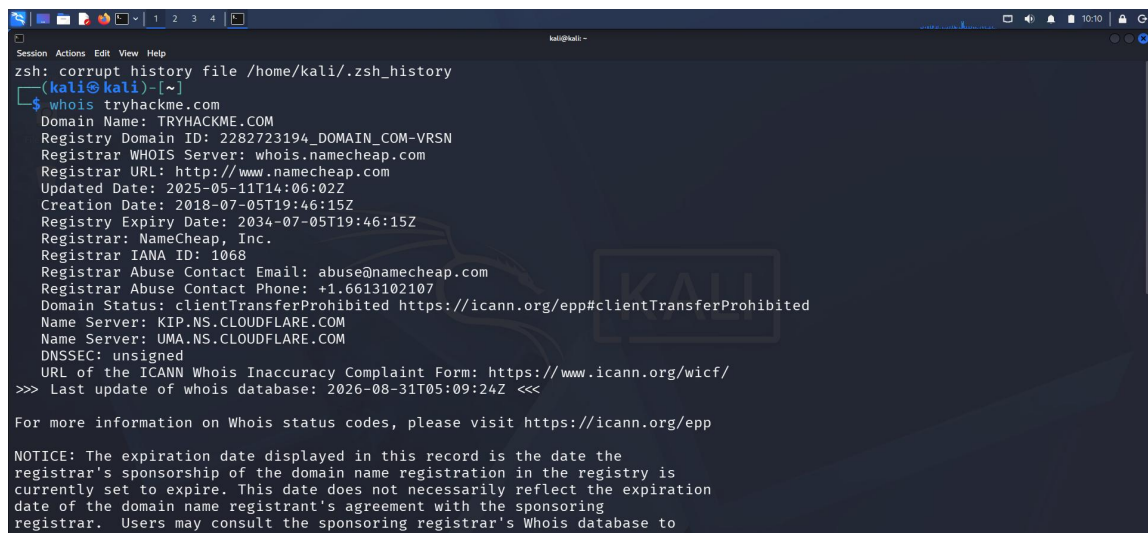
2) Scope

The scope of this task is to collect information without actively engaging with target only gather information that is publicly available using OSINT. All activities were conducted using publicly available resources without directly engaging, scanning, or interacting with the target's network infrastructure.

3) Tools Used

- Kali Linux
- DNS Dumpster
- The Harvester
- Dig
- whois

4) Practical Activity WHOIS Lookup

A screenshot of a Kali Linux terminal window. The terminal shows a command prompt where the user has entered 'whois tryhackme.com'. The output displays detailed domain registration information for 'tryhackme.com', including the domain name, registry ID, registrar (NameCheap, Inc.), creation and expiry dates, and contact information. A large, semi-transparent 'KALI' logo is overlaid on the right side of the terminal output.

```
kali@kali:~$ zsh: corrupt history file /home/kali/.zsh_history
(kali@kali)~$ whois tryhackme.com
Domain Name: TRYHACKME.COM
Registry Domain ID: 2282723194_DOMAIN_COM-VRSN
Registrar WHOIS Server: whois.namecheap.com
Registrar URL: http://www.namecheap.com
Updated Date: 2025-05-11T14:06:02Z
Creation Date: 2018-07-05T19:46:15Z
Registry Expiry Date: 2034-07-05T19:46:15Z
Registrar: NameCheap, Inc.
Registrar IANA ID: 1068
Registrar Abuse Contact Email: abuse@namecheap.com
Registrar Abuse Contact Phone: +1.6613102107
Domain Status: clientTransferProhibited https://icann.org/epp#clientTransferProhibited
Name Server: KIP.NS.CLOUDFLARE.COM
Name Server: UMA.NS.CLOUDFLARE.COM
DNSSEC: unsigned
URL of the ICANN Whois Inaccuracy Complaint Form: https://www.icann.org/wicf/
>>> Last update of whois database: 2026-08-31T05:09:24Z <<<

For more information on Whois status codes, please visit https://icann.org/epp

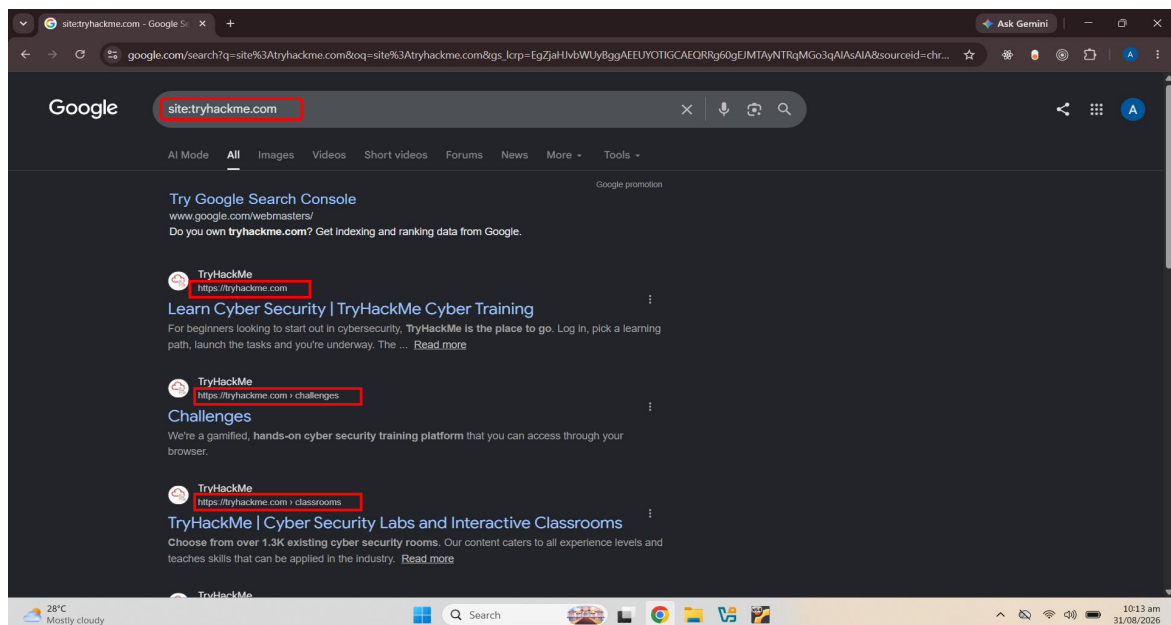
NOTICE: The expiration date displayed in this record is the date the
registrar's sponsorship of the domain name registration in the registry is
currently set to expire. This date does not necessarily reflect the expiration
date of the domain name registrant's agreement with the sponsoring
registrar. Users may consult the sponsoring registrar's Whois database to
```

In the above Figure it is seen that whois lookup is performed on target domain **tryhackme.com**. The information captured via whois lookup is explained below.

Domain Name: TryHackme.com
Registry Domain ID: 2282723194_DOMAIN_COM-VRSN
Registrar whois server: whois.namecheap.com
Registrar URL: <https://www.namecheap.com>
Updated Date: 2025-05-11T14:06:02Z
Creation Date: 2018-07-05T19:46:15Z
Registrar: NameCheap, Inc.
Registrar Abuse Contact Email: abuse@namecheap.com
Registrar Abuse Contact Phone: +1.6613102107

The above information can be utilized by threat actors for a social engineering attack, attempt domain hijacking via registrar supports pretext or align attack timelines around domain renewal window.

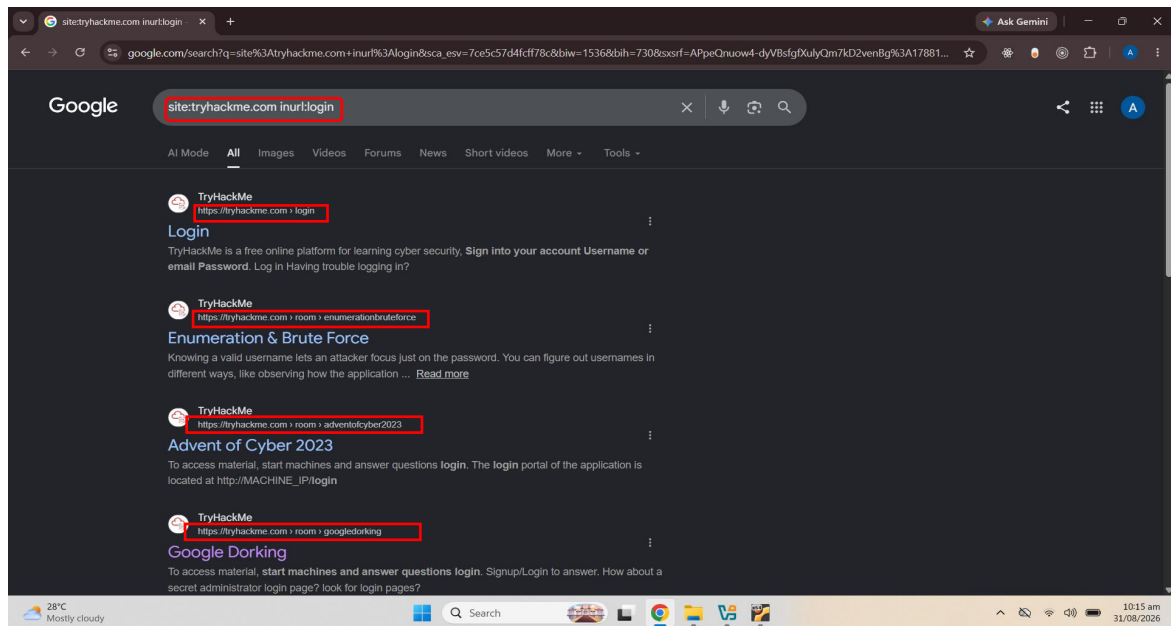
5) Practical Activity Google Dorking



In the above Figure google dorking query **site** is executed against a domain tryhackme.com. The result revealed public facing sub pages and application endpoints that includes

- Main landing Page (tryhackme.com)
- Challenges Portal (<https://tryhackme.com/challenges>)
- Cyber Security Labs and Classrooms(<https://tryhackme.com/classrooms>)

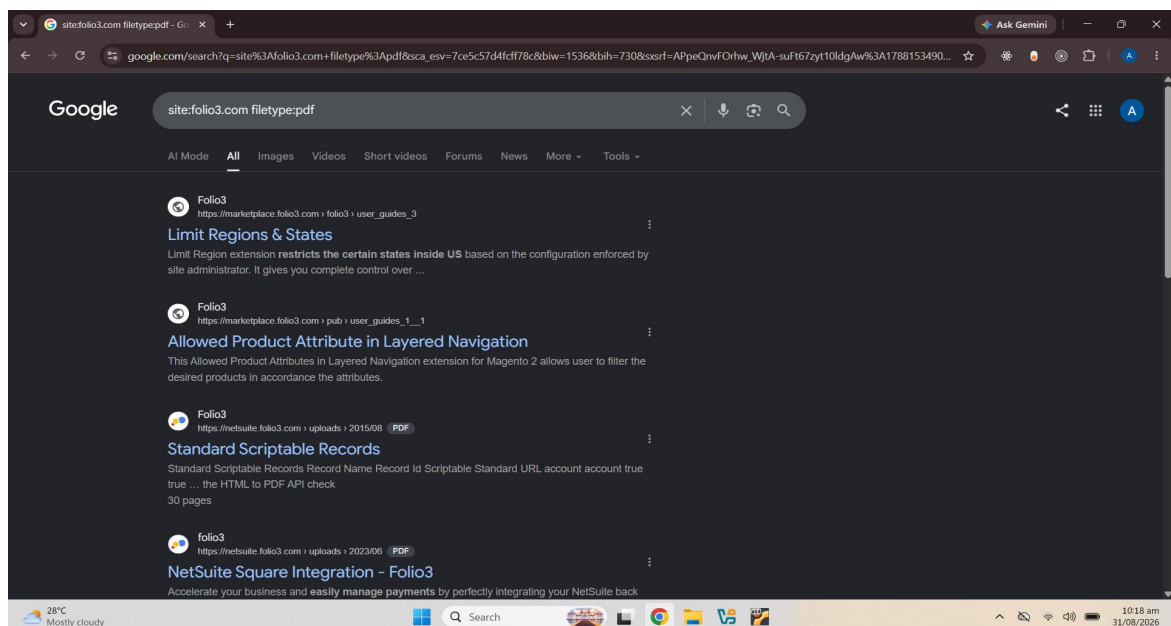
The **site:** operator help threat actors to reveal specific internal directories and functionality hubs as shown in above image (/challenges, /classrooms) giving attackers a clear blue print of available application workflows to target for parameter tampering and authorization bypass .



In the above figure the google dorking query `inurl:login` against the target `tryhackme.com` is executed and the information retrieved is explained below.

- Main Login Portal (<https://tryhackme.com/login>)
- Authentication related resource Specific room pages referencing login mechanics and brute-forcing methodologies. (<https://tryhackme.com/room/enumerationbruteforce>)
- Educational room (<https://tryhackme.com/room/adventcyber2023>)

The `inurl:login` operator allows threat actors to instantly locate the authentication and administration portals which will help threat actors to identify immediate targets for credentials stuffing and brute-force attacks.



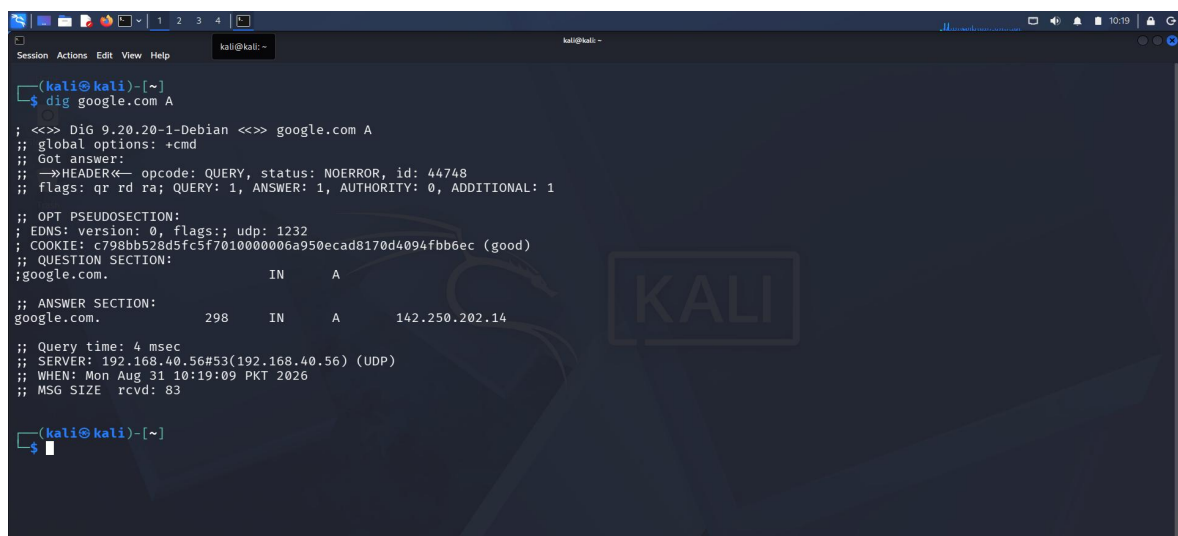
As shown in above figure the google dorking query **site:folio3.com filetype:pdf** was executed against a target domain **folio3.com**. The retrieved information explained below.

- Extension documentation: User guide for Limit Regions & States extension (**marketplace.folio3.com**)
- Product Attribute Guide : User guide for Magento 2 layered navigation (**marketplace.folio3.com**)
- Technical Records Document : Standard Scriptable Records PDF hosted on a NetSuite (**https://netsuite.folio3.com/uploads/2015/08**)
- Integration Documentation: (**https://netsuite.folio3.com/uploads/2023/06**)

The **filetype:pdf** operator isolates publicly accessible documents that often contain sensitive internal intelligence.. Threat actors will utilize this information for targeted spear-phishing attacks.

6) Practical Activity DNS Lookup (A Record)

Command: **dig google.com A**



```
(kali@kali)~$ dig google.com A

; <<>> Dig 9.20.20-1-Debian <<>> google.com A
;; global options: +cmd
;; Got answer:
;;->HEADER<- opcode: QUERY, status: NOERROR, id: 44748
;; flags: qr rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
;; EDNS: version: 0, flags:; udp: 1232
;; COOKIE: c798bb528d5fc5f7010000006a950ecad8170d4094fbb6ec (good)
;; QUESTION SECTION:
;google.com.                IN      A
;; ANSWER SECTION:
google.com.                 298     IN      A      142.250.202.14

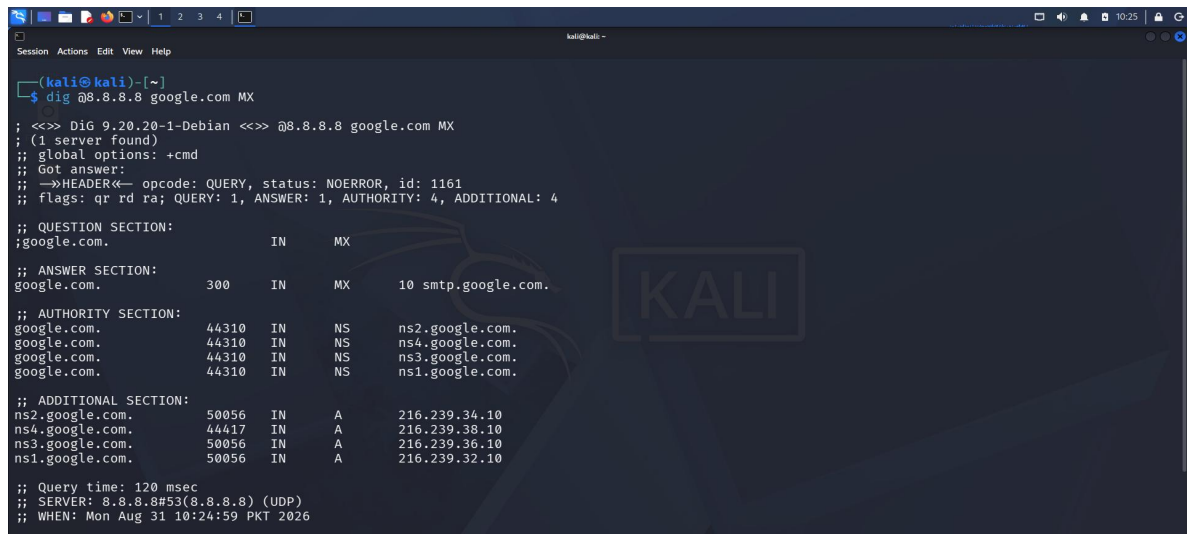
;; Query time: 4 msec
;; SERVER: 192.168.40.56#53(192.168.40.56) (UDP)
;; WHEN: Mon Aug 31 10:19:09 PKT 2026
;; MSG SIZE rcvd: 83

(kali@kali)~$
```

In the above Figure dig command was executed in Kali linux to retrieve the A record of the google domain. The IP address associated is **142.250.202.14**. DNS A records map a human readable domain name directly to its corresponding IP. When some one user or threat actors types google.com into browser the DNS system queries the authoritative name server for domains A record. The DNS server returns IP address enabling network packets to route directly to host machine serving that domain across the internet. The revelation of IP address allows attacker to perform scanning on the target.

7) Practical Activity DNS Lookup (MX Record)

Command: **dig @8.8.8.8 google.com MX**



```
(kali@kali)-[~]
$ dig @8.8.8.8 google.com MX

;<> DiG 9.20.20-1-Debian <> @8.8.8.8 google.com MX
; (1 server found)
;; global options: +cmd
;; Got answer:
;;->HEADER<- opcode: QUERY, status: NOERROR, id: 1161
;; flags: qr rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 4, ADDITIONAL: 4

;; QUESTION SECTION:
;google.com.                IN      MX

;; ANSWER SECTION:
google.com.                 300     IN      MX      10 smtp.google.com.

;; AUTHORITY SECTION:
google.com.                 44310   IN      NS      ns2.google.com.
google.com.                 44310   IN      NS      ns4.google.com.
google.com.                 44310   IN      NS      ns3.google.com.
google.com.                 44310   IN      NS      ns1.google.com.

;; ADDITIONAL SECTION:
ns2.google.com.             50056   IN      A       216.239.34.10
ns4.google.com.             44417   IN      A       216.239.38.10
ns3.google.com.             50056   IN      A       216.239.36.10
ns1.google.com.             50056   IN      A       216.239.32.10

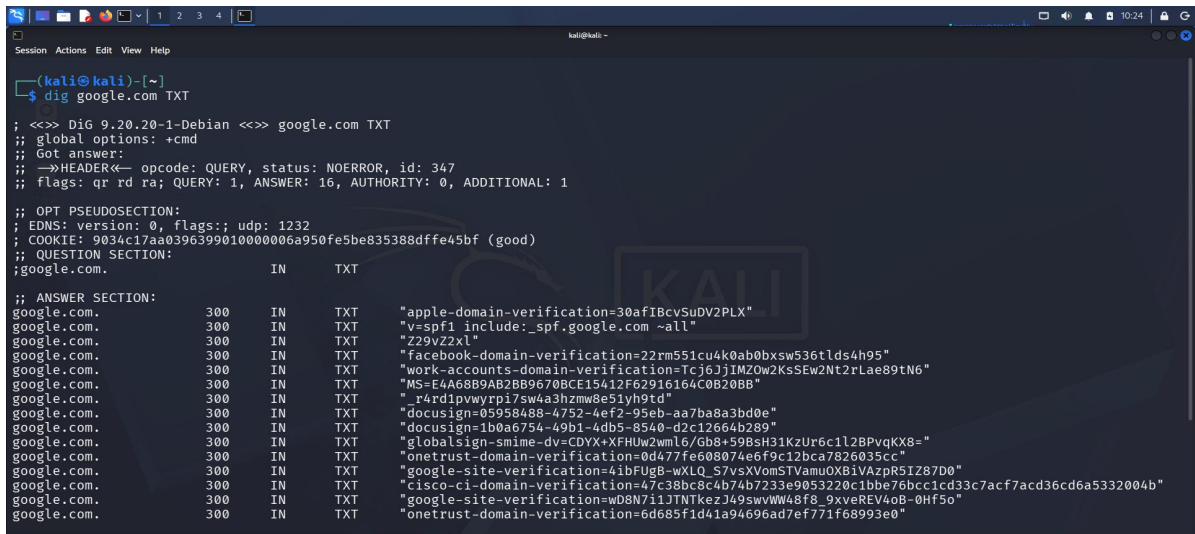
;; Query time: 120 msec
;; SERVER: 8.8.8.8#53(8.8.8.8) (UDP)
;; WHEN: Mon Aug 31 10:24:59 PKT 2026
```

In the above Figure **dig @8.8.8.8 google.com MX** command was executed in Kali linux to retrieve the Mail Exchange (**MX**) record of the google domain. The returned mail server handles inbound email of domain. IP's retrived are

- 216.239.34.10
- 216.239.38.10
- 216.239.36.10
- 216.239.32.10

DNS MX (**Mail Exchange**) records specify the mail servers responsible for accepting incoming email messages on behalf of a domain, along with their relative priority values. Discovering organizations email service infrastructure and third party mail providers. Threat actors use this intelligence to target mail gateways with email spoofing. Eamine mail security controls such as DKIM/SPF/DMARCand map out candaidates for tageted phishing attacks.

8) Practical Activity DNS Lookup (TXT Record)



```
(kali@kali)-[~]
$ dig google.com TXT

; <<>> DiG 9.20.20-1-Debian <<>> google.com TXT
;; global options: +cmd
;; Got answer:
;; ->HEADER<- opcode: QUERY, status: NOERROR, id: 347
;; flags: qr rd ra; QUERY: 1, ANSWER: 16, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1232
; COOKIE: 9034c17aa0396399010000006a950fe5be835388dffe45bf (good)
;; QUESTION SECTION:
;google.com.                IN      TXT

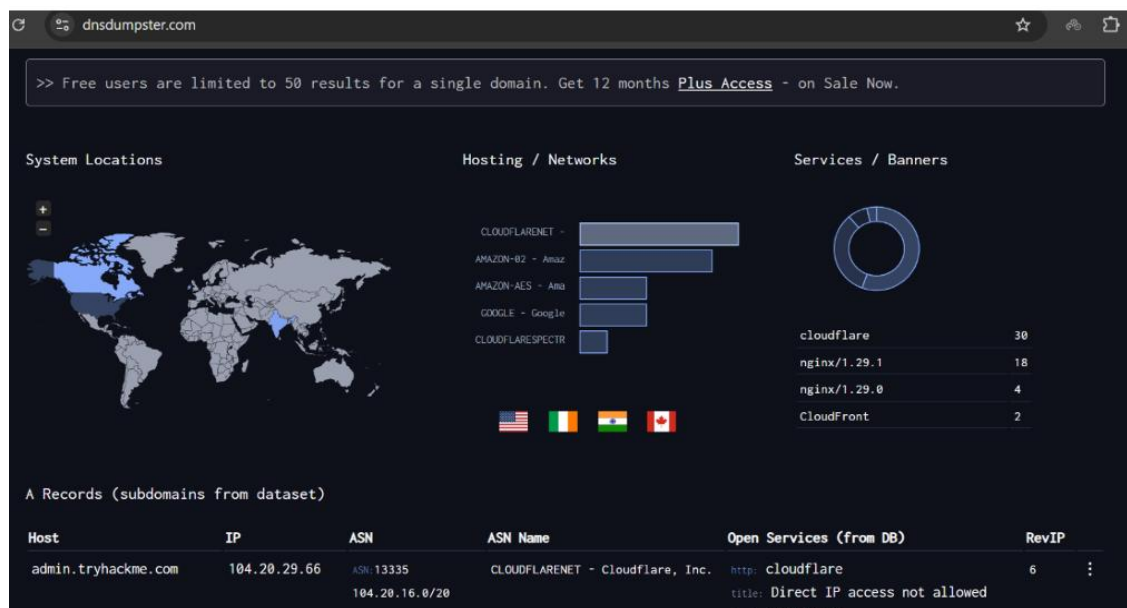
;; ANSWER SECTION:
google.com.                300     IN      TXT     "apple-domain-verification=30aFIBcvSuDV2PLX"
google.com.                300     IN      TXT     "v=spf1 include:_spf.google.com ~all"
google.com.                300     IN      TXT     "Z29vZ2xl"
google.com.                300     IN      TXT     "facebook-domain-verification=22rm551cu4k0ab0bxswS36tlds4h95"
google.com.                300     IN      TXT     "work-accounts-domain-verification=IcJ6JJIMZ0w2KsSEw2Nt2rLae89tN6"
google.com.                300     IN      TXT     "MS=E4A68B9AB2BB9670CE15412F62916164C0B20BB"
google.com.                300     IN      TXT     "r4rd1pvwyrb17swka3hzmw8e51yb9td"
google.com.                300     IN      TXT     "docuSign=05958488-4752-4ef2-95eb-aa7ba8a3bd0e"
google.com.                300     IN      TXT     "docusign=1b0a6754-49b1-4db5-8540-d2c12664b289"
google.com.                300     IN      TXT     "globalsign-smime-dv=CDYX+XFHUw2wml6/Gb8+59BsH31KzUr6c1L2BPvqKX8="
google.com.                300     IN      TXT     "onetrust-domain-verification=0d477fe608074e6f9c12bca7826035cc"
google.com.                300     IN      TXT     "google-site-verification=4ibFUG8-wXLQ_S7vsXVomSTVamu0XBIVazpR5IZ87D0"
google.com.                300     IN      TXT     "cisco-ci-domain-verification=47c38bc8c4b74b7233e9053220c1bbe76bcc1cd33cacf7acd36cd6a5332004b"
google.com.                300     IN      TXT     "google-site-verification=wD8N7i1JNTKczJ49swvWW48f8_9xveREV4o8-0Hf5o"
google.com.                300     IN      TXT     "onetrust-domain-verification=6d685f1d41a94696ad7ef771f68993e0"
```

As shown in above Figure, the **dig google.com TXT** command was executed in Kali Linux to retrieve the Text (TXT) records of the target domain **google.com**. The returned records contain domain verification strings and email authentication configurations, including:

- **Email Authentication (SPF):** v=spf1 include:_spf.google.com ~all
- **Third-Party Service Verifications:** Apple, Facebook, Microsoft, DocuSign, Cisco, OneTrust, and GlobalSign S/MIME domain ownership strings.

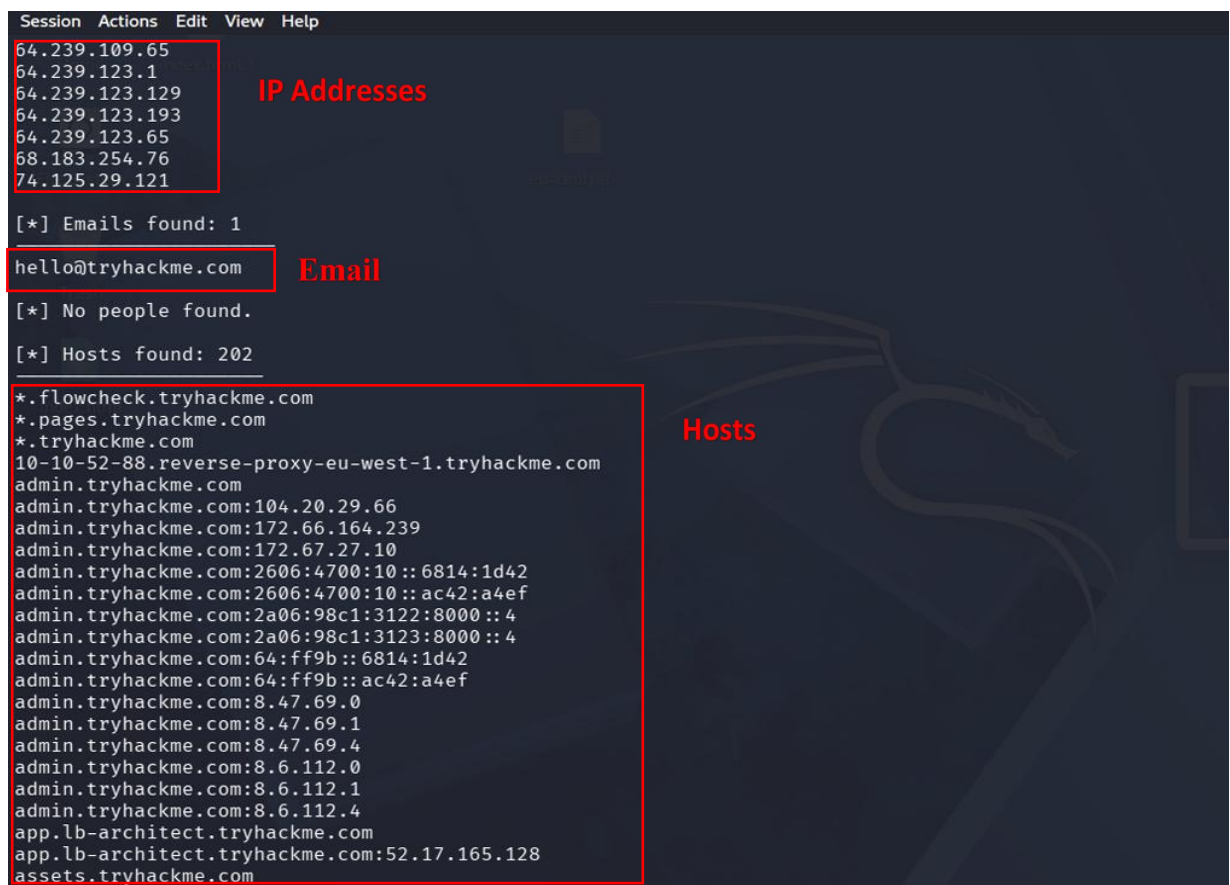
This **TXT** records helps threat actors to Analyze TXT records which reveals an organization's complete third-party service supply chain and email security posture which help Threat actors to leverage this data to identify vendor dependencies for targeted supply-chain phishing, evaluate weaknesses in SPF policies to attempt email spoofing.

9) Practical Activity Subdomain Enumeration using DNSDumpster



In the Above Picture DNSDumpster was used to perform passive subdomain enumeration for the TryHackMe domain. The results displayed discovered subdomains, associated IP addresses, hosting providers, and publicly available DNS information.

10) Practical Activity Information Gathering using theHarvester

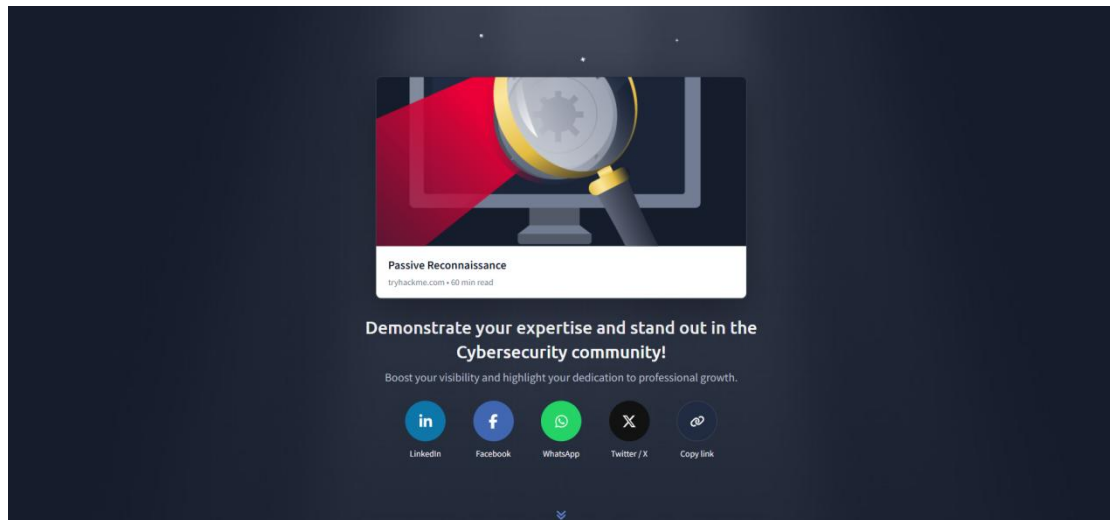


The screenshot displays the theHarvester tool interface with a dark theme. The menu bar at the top includes 'Session', 'Actions', 'Edit', 'View', and 'Help'. The main content area is divided into sections for different types of information gathered. A red box highlights the 'IP Addresses' section, which lists several IP addresses. Another red box highlights the 'Email' section, showing a single email address. A third red box highlights the 'Hosts' section, listing various subdomains and their corresponding IP addresses. The background features a faint, stylized dragon logo.

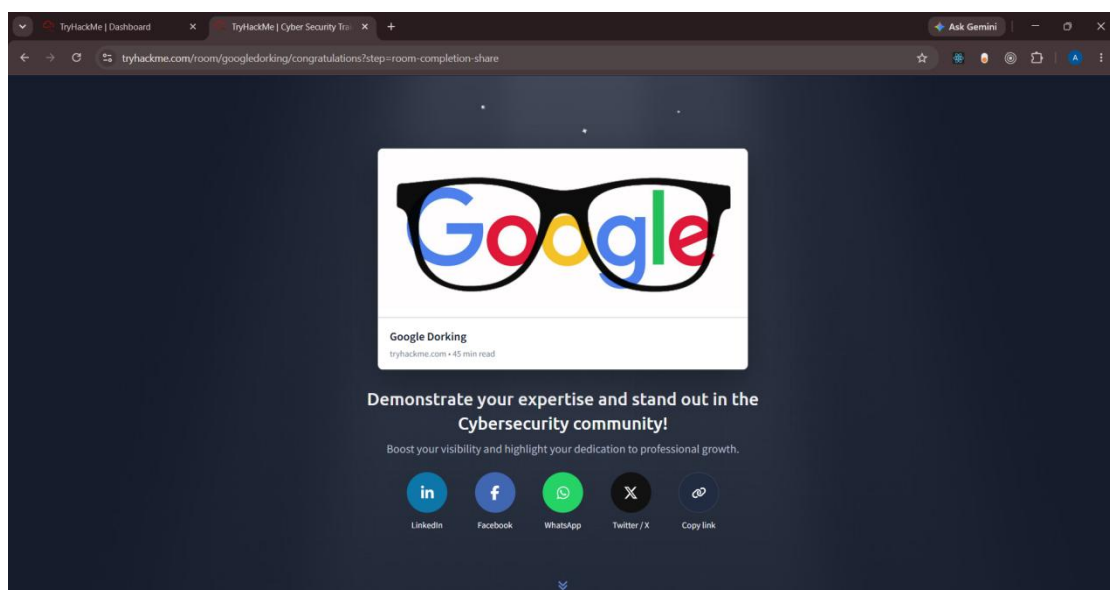
```
Session Actions Edit View Help
64.239.109.65
64.239.123.1
64.239.123.129
64.239.123.193
64.239.123.65
68.183.254.76
74.125.29.121
IP Addresses
[*] Emails found: 1
hello@tryhackme.com
Email
[*] No people found.
[*] Hosts found: 202
*.flowcheck.tryhackme.com
*.pages.tryhackme.com
*.tryhackme.com
10-10-52-88.reverse-proxy-eu-west-1.tryhackme.com
admin.tryhackme.com
admin.tryhackme.com:104.20.29.66
admin.tryhackme.com:172.66.164.239
admin.tryhackme.com:172.67.27.10
admin.tryhackme.com:2606:4700:10::6814:1d42
admin.tryhackme.com:2606:4700:10::ac42:a4ef
admin.tryhackme.com:2a06:98c1:3122:8000::4
admin.tryhackme.com:2a06:98c1:3123:8000::4
admin.tryhackme.com:64:ff9b::6814:1d42
admin.tryhackme.com:64:ff9b::ac42:a4ef
admin.tryhackme.com:8.47.69.0
admin.tryhackme.com:8.47.69.1
admin.tryhackme.com:8.47.69.4
admin.tryhackme.com:8.6.112.0
admin.tryhackme.com:8.6.112.1
admin.tryhackme.com:8.6.112.4
app.lb-architect.tryhackme.com
app.lb-architect.tryhackme.com:52.17.165.128
assets.tryhackme.com
Hosts
```

In the above figure The **theHarvester** tool was used to collect publicly available information related to the tryhackme.com domain, including email addresses, IP addresses, and Hosts.

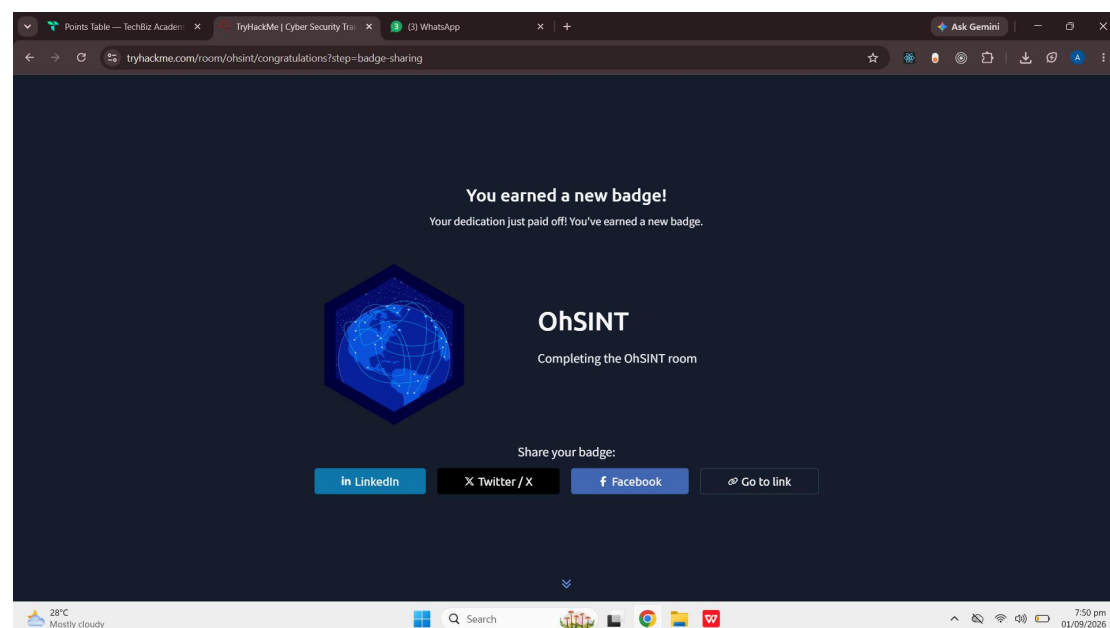
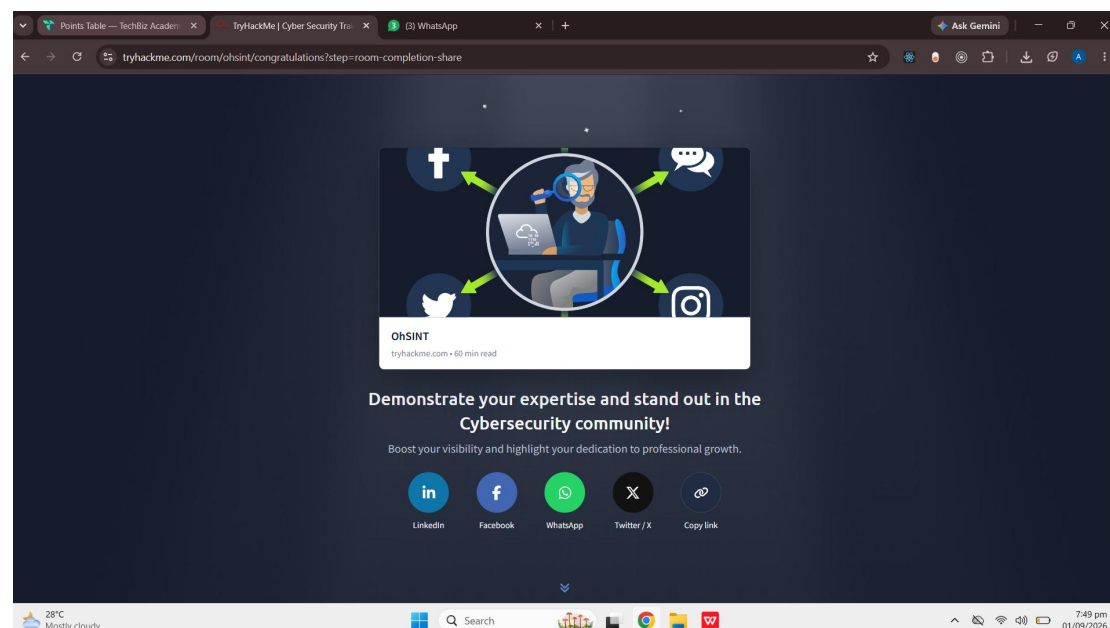
11) TryHackMe Passive Reconnaissance Room



12) TryHackMe Google Dorking Room



13) TryHackMe OhSINT Room



14) Conclusion

This task provided practical experience in passive reconnaissance and OSINT techniques. Using tools such as WHOIS, Google Dorking, DNS lookups, DNSDumpster, and theHarvester, I successfully gathered publicly available information without directly interacting with the target. The TryHackMe rooms enhanced my understanding of information gathering, DNS enumeration, subdomain discovery, and Google search operators. Overall, this task strengthened my knowledge of the reconnaissance phase in ethical hacking.

